Machine Learning 2026 • Tutorial 3 **SOLUTIONS**

Linear Regression: the Mathematical Foundations

21 August 2026 • 08:30–09:50 • pen and paper; no calculator needed

We will move from a sum of squared errors to matrix notation, a gradient, and a projection. The numbers are small so that you can focus on the reasoning. Work through Questions 1–3. We will do Question 4 at the board.

Keep the two spaces apart. $\mathbf{y}, \hat{\mathbf{y}}, \hat{\mathbf{r}}$ and the *columns* of X live in $\mathbb{R}^n$, one slot per data point; $\boldsymbol{\theta}$, ∇J live in $\mathbb{R}^d$, one slot per coefficient. Keep checking the dimensions as you work.

Question 1 From a summation to a vector

20 minutes

A network is tested with different numbers of extra hidden layers. Here x_i is the number of extra layers and y_i is the measured latency. The model

$$f_{\boldsymbol{\theta}}(x) = \theta_0 + \theta_1 x$$

has a fixed overhead θ_0 and an added latency θ_1 per layer. For these four runs, $n = 4$, $d = 2$, and

$$J(\boldsymbol{\theta}) = \tfrac{1}{2}\sum_{i=1}^{n}(y_i - \theta_0 - \theta_1 x_i)^2.$$

extra layers x_i	0	1	2	3
latency y_i (ms)	2	2	4	8

In ML terms: *X and $\mathbf{y}$ contain the data. The parameter vector $\boldsymbol{\theta}$ specifies one model. The loss J scores that model, and $\mathbf{r}$ records its errors.*

(a) Already stacked for you: $X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 2 \\ 2 \\ 4 \\ 8 \end{bmatrix}$. State the shape of X. What does one row represent? What does each column represent? Why is the first column all ones? Finally, write entry i of $X\boldsymbol{\theta}$.

1. $n = 4$ runs, so X has *four rows*; $d = 2$ coefficients, so *two columns*. Write the empty 4×2 frame first.
2. Row i is $[\,1 \;\; x_i\,]$: the 1 is what θ_0 multiplies.
3. $X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 2 \\ 2 \\ 4 \\ 8 \end{bmatrix}$.

(b) Show that entry i of $\mathbf{y} - X\boldsymbol{\theta}$ is $r_i = y_i - \theta_0 - \theta_1 x_i$. Hence show that $J(\boldsymbol{\theta}) = \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\theta}\|_2^2$. Then write $\sum_i r_i$ and $\sum_i x_i r_i$ as dot products. Stack the two answers.

1. Entry i of $X\boldsymbol{\theta}$: row i times $\boldsymbol{\theta}$ is $1 \cdot \theta_0 + x_i \cdot \theta_1$.
2. So entry i of $\mathbf{y} - X\boldsymbol{\theta}$ is $y_i - \theta_0 - \theta_1 x_i$, which is r_i.
3. $\|\mathbf{r}\|_2^2 = \sum_i r_i^2$, so $J(\boldsymbol{\theta}) = \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\theta}\|_2^2$. The norm *is* the sum.
4. $\sum_i r_i = \mathbf{1}^\top \mathbf{r}$ (column 1 of X); $\sum_i x_i r_i = \mathbf{x}^\top \mathbf{r}$ (column 2); $\sum_i r_i^2 = \mathbf{r}^\top \mathbf{r}$.
5. Stack the first two: $X^\top \mathbf{r}$.

(c) Evaluate J at $\boldsymbol{\theta} = (1, 2)^\top$ and at $\boldsymbol{\theta} = (0, 2)^\top$, once as a sum and once as $\frac{1}{2}\|\mathbf{y} - X\boldsymbol{\theta}\|_2^2$.

1. $\boldsymbol{\theta} = (1, 2)$: predictions $1, 3, 5, 7$; residuals $1, -1, -1, 1$; $\sum r_i^2 = 4$; $J = 2$.
2. $\boldsymbol{\theta} = (0, 2)$: predictions $0, 2, 4, 6$; residuals $2, 0, 0, 2$; $\sum r_i^2 = 8$; $J = 4$.
3. Same numbers by either route — they are one expression in two notations.

Beyond regression: *For squared loss, stacking the residuals turns a long summation into one norm. Other losses can also be summed over examples, but they need not be norms.*

Question 2 Transposes: expanding the objective function *22 minutes*

We now expand the loss before differentiating it. Use these three facts: (i) $\|\mathbf{v}\|_2^2 = \mathbf{v}^\top \mathbf{v}$; (ii) $(AB)^\top = B^\top A^\top$, so $(X\boldsymbol{\theta})^\top = \boldsymbol{\theta}^\top X^\top$; (iii) a 1×1 matrix is its own transpose.

In ML terms: *This is the* objective *rearranged into a form we can differentiate. For least squares, the gradient leads both to a closed-form solution and to iterative optimisation methods.*

(a) Multiplying out $\frac{1}{2}(\mathbf{y} - X\boldsymbol{\theta})^\top(\mathbf{y} - X\boldsymbol{\theta})$ gives *four* terms:

$$\tfrac{1}{2}\big[\,\mathbf{y}^\top\mathbf{y} \underbrace{-\,\mathbf{y}^\top X\boldsymbol{\theta}}_{\text{(i)}} \underbrace{-\,\boldsymbol{\theta}^\top X^\top\mathbf{y}}_{\text{(ii)}} + \boldsymbol{\theta}^\top X^\top X\boldsymbol{\theta}\,\big].$$

Show that terms (i) and (ii) are the same single number (a scalar). Combine them and write J using three terms.

1. Write $\|\mathbf{v}\|_2^2 = \mathbf{v}^\top\mathbf{v}$ with $\mathbf{v} = \mathbf{y} - X\boldsymbol{\theta}$: $J = \frac{1}{2}(\mathbf{y} - X\boldsymbol{\theta})^\top(\mathbf{y} - X\boldsymbol{\theta})$.
2. Multiply out as $(a-b)^\top(a-b)$, keeping the order — *four* terms:
$$\tfrac{1}{2}\big[\mathbf{y}^\top\mathbf{y} - \mathbf{y}^\top X\boldsymbol{\theta} - \boldsymbol{\theta}^\top X^\top\mathbf{y} + \boldsymbol{\theta}^\top X^\top X\boldsymbol{\theta}\big].$$
3. The middle terms are scalars. Count shapes for $\mathbf{y}^\top X\boldsymbol{\theta}$: $(1\times n)(n\times d)(d\times 1) = 1\times 1$. It is a *number*.
4. A number equals its own transpose. So $\mathbf{y}^\top X\boldsymbol{\theta} = (\mathbf{y}^\top X\boldsymbol{\theta})^\top$.
5. Apply $(AB)^\top = B^\top A^\top$ twice: $(\mathbf{y}^\top X\boldsymbol{\theta})^\top = \boldsymbol{\theta}^\top X^\top\mathbf{y}$.
6. Therefore the two middle terms are the *same number*; together they give $-\boldsymbol{\theta}^\top X^\top\mathbf{y}$ after the $\frac{1}{2}$:
$$\boxed{J(\boldsymbol{\theta}) = \tfrac{1}{2}\mathbf{y}^\top\mathbf{y} - \boldsymbol{\theta}^\top X^\top\mathbf{y} + \tfrac{1}{2}\boldsymbol{\theta}^\top X^\top X\boldsymbol{\theta}}$$
7. Label the three terms: constant in $\boldsymbol{\theta}$, linear in $\boldsymbol{\theta}$, quadratic in $\boldsymbol{\theta}$. That labelling is what makes the next part routine.

(b) Work in three stages:

1. Show $\nabla_{\boldsymbol{\theta}}(\mathbf{a}^\top\boldsymbol{\theta}) = \mathbf{a}$.
2. For a 2×2 matrix A, show $\nabla_{\boldsymbol{\theta}}(\boldsymbol{\theta}^\top A\boldsymbol{\theta}) = (A + A^\top)\boldsymbol{\theta}$.
3. Use these rules to find ∇J. Remember that $X^\top X$ is symmetric.

1. $\mathbf{a}^\top\boldsymbol{\theta} = a_1\theta_1 + a_2\theta_2$. Partials: a_1 and a_2. Stack: $\nabla_{\boldsymbol{\theta}}(\mathbf{a}^\top\boldsymbol{\theta}) = \mathbf{a}$. (Scalar analogue: $\mathrm{d}(ax)/\mathrm{d}x = a$.)
2. $\boldsymbol{\theta}^\top A\boldsymbol{\theta} = A_{11}\theta_1^2 + (A_{12} + A_{21})\theta_1\theta_2 + A_{22}\theta_2^2$.
3. Partials: $2A_{11}\theta_1 + (A_{12} + A_{21})\theta_2$ and $(A_{12} + A_{21})\theta_1 + 2A_{22}\theta_2$; stacked, that is $(A + A^\top)\boldsymbol{\theta}$. Symmetric A: $2A\boldsymbol{\theta}$.
4. Now go term by term through part (a). Constant term: contributes nothing.
5. Linear term $-\boldsymbol{\theta}^\top X^\top\mathbf{y}$: by step 1 its gradient is $-X^\top\mathbf{y}$.
6. Quadratic term $\frac{1}{2}\boldsymbol{\theta}^\top X^\top X\boldsymbol{\theta}$: $X^\top X$ is symmetric, so by step 3 its gradient is $\frac{1}{2}\cdot 2X^\top X\boldsymbol{\theta} = X^\top X\boldsymbol{\theta}$.
7. Add: $\nabla J(\boldsymbol{\theta}) = X^\top X\boldsymbol{\theta} - X^\top\mathbf{y} = X^\top(X\boldsymbol{\theta} - \mathbf{y}) = -X^\top\mathbf{r}$.

Beyond regression: *These rules generate every gradient you will derive from here on; backpropagation is this same chain rule, applied one layer at a time.*

Question 3 Solve the normal equation; read the geometry *20 minutes*

You found $\nabla J(\boldsymbol{\theta}) = -X^\top \mathbf{r}$. For this convex least-squares problem, the minimum satisfies $\nabla J = \mathbf{0}$. We now solve this equation and then read its geometric meaning.

In ML terms: *At the least-squares solution, $X^\top \widehat{\mathbf{r}} = \mathbf{0}$. In $\mathbb{R}^n$ this says that the remaining residual is perpendicular to every column of X. There is no remaining* linear *correction available within this model space.*

(a) The **normal equation** for least squares is

$$\boxed{X^\top X \widehat{\boldsymbol{\theta}} = X^\top \mathbf{y}}.$$

For the runs in Question 1, $X^\top X = \left[\begin{smallmatrix} 4 & 6 \\ 6 & 14 \end{smallmatrix}\right]$ and $X^\top \mathbf{y} = \left[\begin{smallmatrix} 16 \\ 34 \end{smallmatrix}\right]$ — check one entry of each, then use

$$\begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} \widehat{\theta}_0 \\ \widehat{\theta}_1 \end{bmatrix} = \begin{bmatrix} 16 \\ 34 \end{bmatrix}.$$

Thus

$$4\widehat{\theta}_0 + 6\widehat{\theta}_1 = 16, \qquad 6\widehat{\theta}_0 + 14\widehat{\theta}_1 = 34.$$

Solve these equations. What do the two fitted coefficients mean?

1. $X^\top X = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}$. The four sums are: 4, 0+1+2+3 = 6, 0+1+4+9 = 14.
2. $X^\top \mathbf{y} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix} = \begin{bmatrix} 16 \\ 34 \end{bmatrix}$, since $\sum x_i y_i = 0 + 2 + 8 + 24 = 34$.
3. The normal equations: $4\theta_0 + 6\theta_1 = 16$ and $6\theta_0 + 14\theta_1 = 34$.
4. From the first, $\theta_0 = 4 - 1.5\theta_1$. Substitute: $24 - 9\theta_1 + 14\theta_1 = 34$, so $5\theta_1 = 10$.
5. $\theta_1 = 2$, $\theta_0 = 1$: $\widehat{\boldsymbol{\theta}} = (1, 2)^\top$.
6. Read it back: **1 ms of fixed overhead, 2 ms per added layer.**

(b) Your fitted values are $\widehat{\mathbf{y}} = (1, 3, 5, 7)^\top$.

1. Find $\widehat{\mathbf{r}} = \mathbf{y} - \widehat{\mathbf{y}}$.
2. Compute $X^\top \widehat{\mathbf{r}}$. What do its two zeros mean?
3. Verify $\|\mathbf{y}\|_2^2 = \|\widehat{\mathbf{y}}\|_2^2 + \|\widehat{\mathbf{r}}\|_2^2$.

1. $\widehat{\mathbf{y}} = X\widehat{\boldsymbol{\theta}} = (1, 3, 5, 7)^\top$.
2. $\widehat{\mathbf{r}} = \mathbf{y} - \widehat{\mathbf{y}} = (2, 2, 4, 8) - (1, 3, 5, 7) = (1, -1, -1, 1)^\top$.
3. $X^\top \widehat{\mathbf{r}}$: first entry $1 - 1 - 1 + 1 = 0$; second entry $0 - 1 - 2 + 3 = 0$. Both zero.
4. Meaning: $\widehat{\mathbf{r}}$ is perpendicular to *each column* of X, hence to everything they span — $\widehat{\mathbf{r}} \perp \mathrm{Col}(X)$.
5. Count them: this is d equations, one per column, not one.
6. Norms: $\|\mathbf{y}\|^2 = 4+4+16+64 = 88$; $\|\widehat{\mathbf{y}}\|^2 = 1+9+25+49 = 84$; $\|\widehat{\mathbf{r}}\|^2 = 4$. And $84 + 4 = 88$.
7. That is Pythagoras, and it holds only because $\widehat{\mathbf{y}} \perp \widehat{\mathbf{r}}$.

Beyond regression: *For differentiable convex objectives, a zero gradient identifies a global minimum. When a convenient closed form is unavailable, an optimiser can use the gradient iteratively.*

Question 4 When two features say almost the same thing *12 minutes, at the board*

A model may predict well while its individual coefficients remain unstable. This happens when two feature columns carry almost the same information. We will see it first with numbers, then with identical columns.

Two teammates log capacity separately — one records *layers*, the other *blocks* — giving columns $\mathrm{f}_1 = (1,2,3,4)^\top$ and $\mathrm{f}_2 = (1,2,3,5)^\top$, differing only on the run where one counted the stem. Both go in as features, latency $\mathbf{y} = (3,5,5,9)^\top$, $n = 4$, $d = 2$, no intercept.

In ML terms: ***Identifiability*** *asks whether the data can determine one parameter vector. Accurate predictions do not guarantee stable coefficients.*

(a) *Worked at the board.* Here $X^\top X = \left[\begin{smallmatrix} 30 & 34 \\ 34 & 39 \end{smallmatrix}\right]$, $\det = 14$, and the two fits are

$$\mathbf{y} = (3,5,5,9)^\top \Rightarrow \widehat{\boldsymbol{\theta}} = (1,1)^\top, \qquad \mathbf{y} = (3,5,5,10)^\top \Rightarrow \widehat{\boldsymbol{\theta}} = (0,2)^\top.$$

Only the last response changed, from 9 to 10. First find the change in $\widehat{\boldsymbol{\theta}}$. Then compute the residual for each fit. Did the quality of the fit change?

1. The coefficient change is $(0,2)^\top - (1,1)^\top = (-1,1)^\top$.
2. First fit: $X(1,1)^\top = (2,4,6,9)^\top$, so $\mathbf{r} = (1,1,-1,0)^\top$.
3. Second fit: $X(0,2)^\top = (2,4,6,10)^\top$, so the residual is again $(1,1,-1,0)^\top$. Both fits therefore have the same squared error, 3.

(b) Now make the two feature columns identical: $\mathrm{f}_1 = \mathrm{f}_2 = (1,2,3,4)^\top$. Then $X^\top X = \left[\begin{smallmatrix} 30 & 30 \\ 30 & 30 \end{smallmatrix}\right]$.

Work through these three short steps:

1. Compute $\det(X^\top X)$.
2. For $\mathbf{v} = (1,-1)^\top$, verify that $X\mathbf{v} = \mathbf{0}$.
3. Use $X(\boldsymbol{\theta} + t\mathbf{v}) = X\boldsymbol{\theta} + tX\mathbf{v}$ to compare the predictions from $\boldsymbol{\theta}$ and $\boldsymbol{\theta} + t\mathbf{v}$. What does this tell you about the fitted coefficients?

1. $X^\top X = \begin{bmatrix} 30 & 30 \\ 30 & 30 \end{bmatrix}$, and $\det = 900 - 900 = 0$.
2. Find $\mathbf{v}$ with $X\mathbf{v} = \mathbf{0}$: the columns are equal, so $\mathbf{v} = (1,-1)^\top$ works. Check all four rows give 0.
3. Predictions: $X(\boldsymbol{\theta} + t\mathbf{v}) = X\boldsymbol{\theta} + tX\mathbf{v} = X\boldsymbol{\theta}$. *Identical*, for every t. So the residual is identical, and so is the loss — flat along $\mathbf{v}$, exactly, not approximately.
4. Therefore the fitted coefficient vector is *not unique*: infinitely many parameter vectors are equally good, and no training procedure can prefer one.

Beyond regression: *The next lecture will connect this invisible parameter direction to rank, eigenvalues, and curvature.*